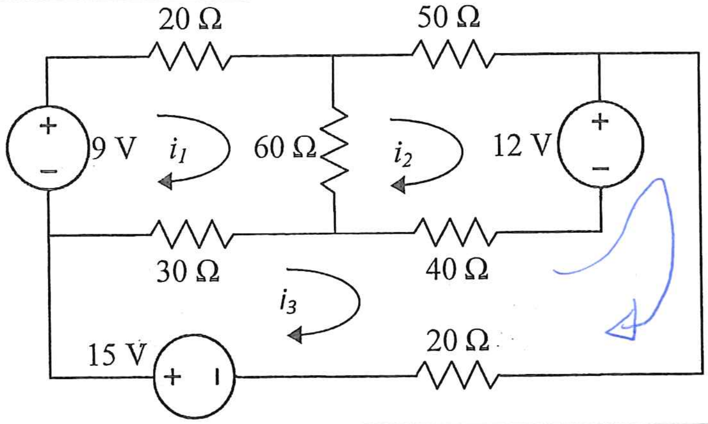
Memo

EBN111 CLASS TEST 2 / KLASTOETS 2
8 April 2016
[30 min / 15 marks]

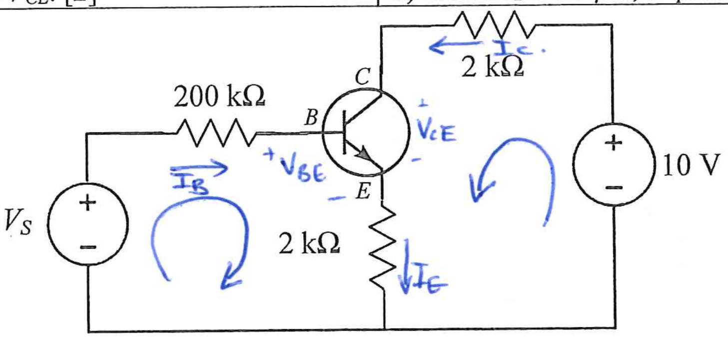
Question 1 [3] Determine the mesh-current equations by means of inspection.	Vraag 1 [3] Bepaal die lus-stroom vergelykings deur inspeksie.



$$\begin{bmatrix} 110 & -60 & -30 \\ -60 & 150 & -40 \\ -30 & -40 & 90 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} = \begin{bmatrix} 9 \\ -12 \\ 27 \end{bmatrix} \begin{matrix} \checkmark \\ \checkmark \\ \checkmark \end{matrix}$$

* 1 mark per correct row

Question 2 [4]	Vraag 2 [4]
Given: $\beta = 99$ and $V_{BE} = 0.7$ V. a) If $I_C = 0.5$ mA, determine V_S. [2] b) If $I_B = 10$ μA, determine V_{CE}. [2]	Gegee: $\beta = 99$ en $V_{BE} = 0.7$ V. a) Indien $I_C = 0.5$ mA, bepaal V_S. [2] b) Indien $I_B = 10$ μA, bepaal V_{CE}. [2]



a) $I_C = 0{,}5mA$

$\therefore I_B = \frac{I_C}{\beta} = 5{,}0505\ \mu A$

$I_E = I_C + I_B = (1+\beta)I_B = 5{,}051 \times 10^{-4} A$

$\therefore -V_S + 200k\, I_B + V_{BE} + 2k\, I_E = 0$

$V_S = 200k(5{,}0505\mu) + 0{,}7 + 2k(5{,}051 \times 10^{-4})$

$= +2{,}72\ V$

✓ ✓

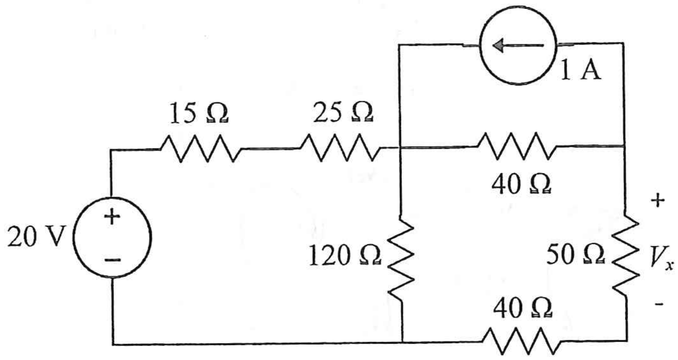
b) $I_B = 10\mu A$

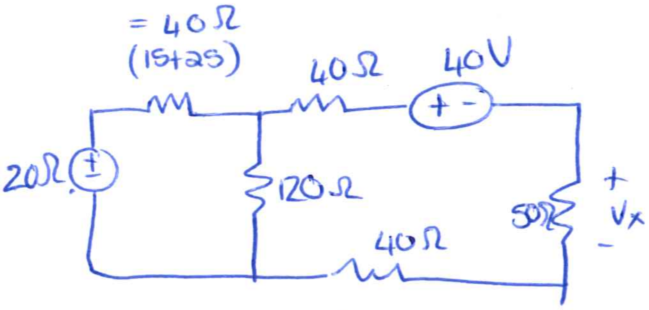
$\therefore -10 + 2k(\beta I_B) + V_{CE} + 2k(1+\beta)I_B = 0$

$V_{CE} = +6{,}02\ V$

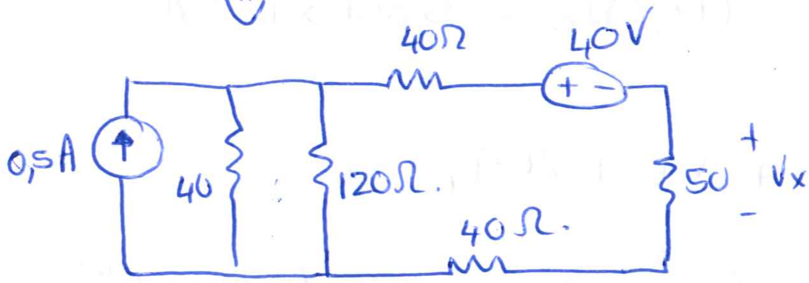
✓ ✓

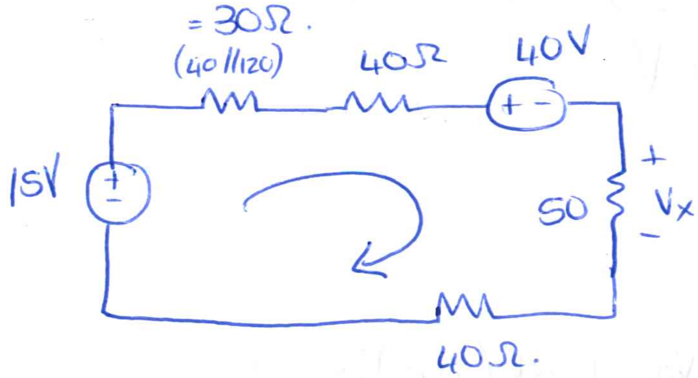
Question 3 [2] Use source transformation and determine V_x.	**Vraag 3 [2]** Gebruik brontransformasie en bepaal V_x.





⇓





$\therefore -15 + (30+40+50+40)I + 40 = 0$

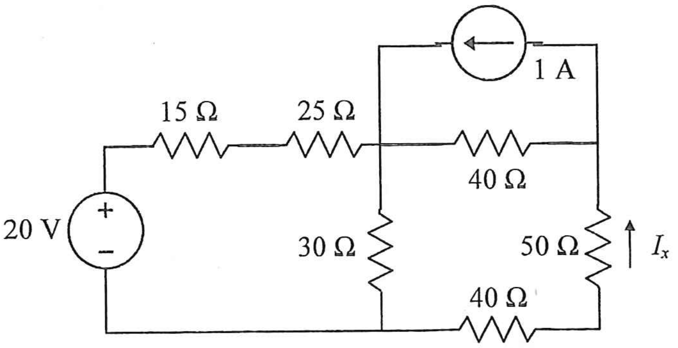
$\therefore I = -0{,}15625 \text{ A}$

$\therefore V_x = I(50)$

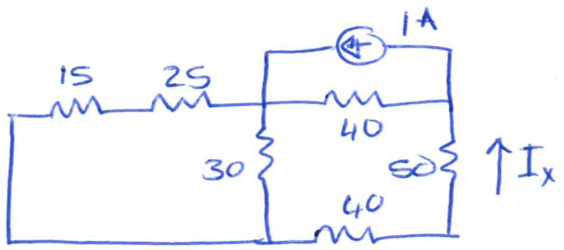
$= -7{,}81 \text{ V}$

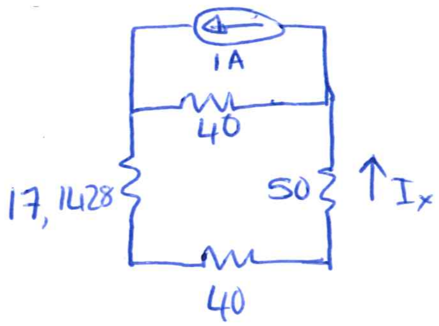
✓ ✓

Question 4 [2]	Vraag 4 [2]
Use superposition and determine I_x due to the current source.	Gebruik superposisie en bereken I_x agv die stroombron.



∴ I_x due to current source:

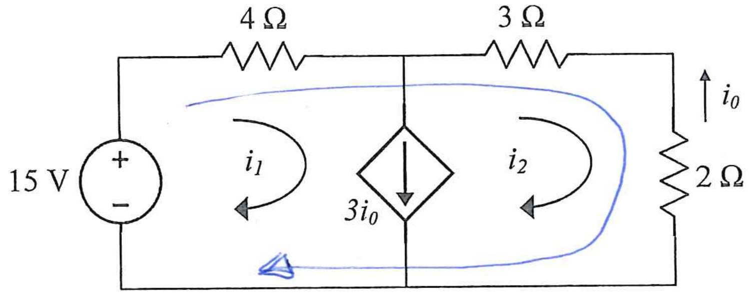




$$\therefore I_x = 1\left(\frac{40}{40+(17,143+40+50)}\right)$$

$$= 0,271 \text{ A}$$

Question 5 [4] Gebruik superlus-analise om twee vergelykings van die volgende vorm op te stel: $ai_1 + bi_2 = 15$ $ci_1 + di_2 = 0$	**Vraag 5 [4]** Use supermesh analysis to set up two equations of the following form: $ai_1 + bi_2 = 15$ $ci_1 + di_2 = 0$



① $-15 + 4i_1 + 3i_2 + 2i_2 = 0$

$4i_1 + 5i_2 = 15$

② $* \; i_0 = -i_2$

$i_1 - i_2 = 3i_0$

$i_1 - i_2 = 3(-i_2)$

$\therefore 1i_1 + 2i_2 = 0$

$a = 4$ ✓ $b = 5$ ✓

$* \; c = 1$ ✓ $d = 2$ ✓

we also accepted

$\boxed{c = -1 \quad d = -2}$

ONLY if both coefficients were negative.

Memo

EBN111 CLASS TEST 2 / KLASTOETS 2 (8 APR 2012)
ANSWER SHEET / ANTWOORDBLAD

Surname: Van:		*Student number:* Studentenommer:								
Initials: Voorletters:		*Group:* Groep:	ENG			ENG (Mech)			AFR	

Tot:	**/15**

1	$$\begin{bmatrix} 110 & -60 & -30 \\ -60 & 150 & -40 \\ -30 & -40 & 90 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} = \begin{bmatrix} 9 \\ -12 \\ 27 \end{bmatrix}$$ ✓ ✓ ✓	
2	*a)* $V_S =$ 2,72V ✓✓	
	b) $V_{CE} =$ 6,02V ✓✓	
3	$V_x =$ −7,81V ✓✓	
4	$I_x =$ 0,271 A ✓✓	
5	a = 4 ✓	b = 5 ✓
	c = 1 ✓	d = 2 ✓